Example Problem on Understeer coefficient

Find the understeer gradient and effective slip angle for a vehicle which is having a weight of 2500 kg and CG is located at 1400 from Front with wheelbase of 2500 mm if the Vehicle is taking a turn of Radius 20 m and at a velocity of 25 kmph .
The Cornering stiffness of Front Tyre is xx N/deg , cornering stiffness of Rear Tyre is

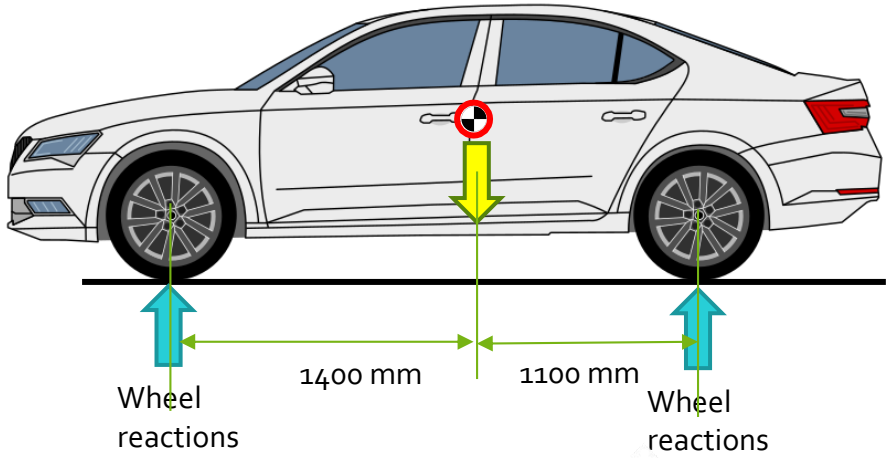
Whether the Car will understeer or Understeer by virtue of its weight distribution ?
What is Effective slip angle if Lateral acceleration is 0.2 g

Understeer gradient measurements by constant radius method

$$W = 2500 * 9.81 = 24525\ N$$

$$W_F = W.\frac{b}{L}$$

$$W_R = W.\frac{a}{L}$$



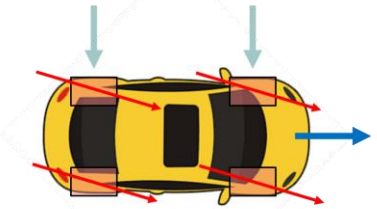
$$W_F = 24525.\frac{1.1}{2.5} = 10791\ N$$

$$W_R = 24525.\frac{1.4}{2.5} = 13734$$

$$C_F = 800\ N/\deg of\ slip\ angle$$

$$C_R = 900\ N/\deg of\ slip\ angle$$

$$Understeer\ gradient, K = \frac{W_F}{C_F} - \frac{W_R}{C_R}$$



$$Understeer\ gradient, K = \frac{10791}{800} - \frac{13734}{900}$$

$$K = -1.77$$

$$Effective\ slip\ angle = (\frac{W_F}{C_F} - \frac{W_R}{C_R})(A_y)$$

Since the Understeer Gradient is Highly negative , the vehicle will tend to oversteer by virtue of weight distribution and Tire property